

THQ #4

By: Ethan Le :D

Question 1

Objective Function:

$$\mathbf{\min} \ 2x_1 + 4x_2 - 4x_3 + 7x_4$$

Constraints:

- $8x_1 - 2x_2 + x_3 - x_4 \leq 50$
- $3x_1 + 5x_2 + 2x_4 \leq 150$
- $x_1 - x_2 + 2x_3 - 4x_4 \leq 100$
- $x_1, x_2, x_3, x_4 \geq 0$ **(non-negative)**

The first step is to add slack variables s_1 , s_2 , and s_3 so everything is in canonical form. Since each constraint is already a $\leq$ inequality with a positive right-hand side, I add a non-negative slack to each equation (no surplus or flipping signs needed).

Canonical Form:

Objective Function:

$$\mathbf{\min} \ 2x_1 + 4x_2 - 4x_3 + 7x_4$$

Constraints:

- $8x_1 - 2x_2 + x_3 - x_4 + s_1 = 50$
- $3x_1 + 5x_2 + 2x_4 + s_2 = 150$
- $x_1 - x_2 + 2x_3 - 4x_4 + s_3 = 100$
- $x_1, x_2, x_3, x_4, s_1, s_2, s_3 \geq 0$ **(non-negative)**

Simplex Algorithm:

$$110011$$

	x_1	x_2	x_3	x_4	s_1	s_2	s_3	RHS
s_1	8	-2	(1)	-1	1	0	0	(50)
s_2	3	5	0	2	0	1	0	150
s_3	1	-1	2	-4	0	0	1	100
	2	4	(-4)	7	0	0	0	0

Enter x_3
 Remove s_1

	x_1	x_2	x_3	x_4	s_1	s_2	s_3	RHS
x_3	8	-2	1	-1	0	0	0	50
s_2	3	5	0	2	0	1	0	150
s_3	-15	7	0	-2	-2	0	1	(0)
	34	(-4)	0	3	4	0	0	200

Enter x_2 & remove s_3

	x_1	x_2	x_3	x_4	s_1	s_2	s_3	RHS
x_3	-2	0	1	$-\frac{1}{5}$	$-\frac{1}{3}$	0	$\frac{2}{3}$	50
s_2	28	0	0	$\frac{16}{5}$	$\frac{10}{3}$	1	$-\frac{15}{3}$	150
x_2	-5	1	0	$-\frac{2}{5}$	$-\frac{2}{3}$	0	$-\frac{1}{3}$	0
	14	0	0	$-\frac{7}{5}$	$\frac{14}{3}$	0	$\frac{14}{3}$	200

Solution:

Optimal minimum is -200 at $(x_1, x_2, x_3, x_4, s_1, s_2, s_3) = (0, 0, 50, 0, 0, 150, 0)$.

Question 2**Objective Function:**

$$\mathbf{max} \ 9x_2 + 2x_3 - x_5$$

Constraints:

- $x_1 - 3x_2 - 4x_4 + 2x_6 = 60$
- $2x_2 - x_4 - x_5 + 4x_6 = -20$
- $x_2 + x_3 + 3x_6 = 10$
- $x_1, x_2, x_3, x_4, x_5, x_6 \geq 0$ **(non-negative)**

To match the min approach from class, I negate the objective so maximizing $9x_2 + 2x_3 - x_5$ becomes minimizing $-9x_2 - 2x_3 + x_5$. Since every constraint is an equality, I add artificial variables r_1 , r_2 , and r_3 as the initial basic variables. I also multiply the second constraint by -1 so the right-hand side is positive. Using the Two-Phase method, Phase I minimizes the sum of the artificials to drive them out of the basis; Phase II then optimizes the original (negated) objective.

Phase I:**Objective Function:**

$$\mathbf{min} \ w = r_1 + r_2 + r_3$$

Constraints:

- $x_1 - 3x_2 - 4x_4 + 2x_6 + r_1 = 60$
- $-2x_2 + x_4 + x_5 - 4x_6 + r_2 = 20$
- $x_2 + x_3 + 3x_6 + r_3 = 10$
- $x_1, x_2, x_3, x_4, x_5, x_6, r_1, r_2, r_3 \geq 0$ **(non-negative)**

Initial BVs: $r_1 = 60$, $r_2 = 20$, $r_3 = 10$ (all other variables = 0).

Phase II:

Once Phase I reaches $\boldsymbol{w} = \mathbf{0}$ with all artificials nonbasic, drop the $\boldsymbol{r}_i$ columns and optimize:

$$\mathbf{min} \ z = -9x_2 - 2x_3 + x_5$$

Simplex Algorithm:

Phase I Tableaus:

4 rows

	x_1	x_2	x_3	x_4	x_5	x_6	r_1	r_2	r_3	RHS
r_1	1	-3	0	-4	0	2	1	0	0	60
r_2	0	-2	0	1	1	-4	0	1	0	20
r_3	0	1	1	0	0	3	0	0	1	10
z	0	-9	-2	0	1	0	0	0	0	0
w	0	0	0	0	0	0	1	1	1	0

↓

	x_1	x_2	x_3	x_4	x_5	x_6	r_1	r_2	r_3	RHS
r_1	1	-3	0	-4	0	2				60
r_2	0	-2	0	1	1	-4				20
r_3	0	1	1	0	0	3				10
z	0	-9	-2	0	1	0				0
w	-1	9	-1	3	-1	-1				-90

Enter x_6 , Remove R_3

	x_1	x_2	x_3	x_4	x_5	x_6	r_1	r_2	r_3	RHS
r_1	1	$-\frac{11}{3}$	$-\frac{2}{3}$	-4	0	0				$\frac{160}{3}$
r_2	0	$-\frac{2}{3}$	$\frac{4}{3}$	1	1	0				$\frac{140}{3}$
x_6	0	$\frac{1}{3}$	$\frac{1}{3}$	0	0	1				$\frac{10}{3}$
z	0	-9	-2	0	1	0				0
w	-1	$\frac{13}{3}$	$\frac{2}{3}$	3	-1	0				$-\frac{260}{3}$

Enter x_1 , Remove R_1

	x_1	x_2	x_3	x_4	x_5	x_6	r_1	r_2	r_3	RHS
x_1	1	$-\frac{11}{3}$	$-\frac{2}{3}$	-4	0	0				$\frac{160}{3}$
r_2	0	$-\frac{2}{3}$	$\frac{4}{3}$	1	1	0				$\frac{140}{3}$
x_6	0	$\frac{1}{3}$	$\frac{1}{3}$	0	0	1				$\frac{10}{3}$
z	0	-9	-2	0	1	0				0
w	0	$\frac{2}{3}$	$-\frac{4}{3}$	-1	-1	0				$-\frac{100}{3}$

Enter x_3
Remove x_6

TYPO

Phase II Tableaus:

							RHS
x_1	1	-3	0	-4	0	2	60
r_2	0	-2	0	1	0	-4	20
x_3	0	1	1	0	0	3	10
Z	0	-7	0	0	1	6	20
w	0	2	0	-1	0	4	-20

Enter x_5 , remove r_2

x_1	1	-3	0	-4	0	2	60
x_5	0	-2	0	1	1	-4	20
x_3	0	1	1	0	0	3	10
Z	0	-7	0	0	1	10	0
w	0	0	0	0	0	0	0

Remove x_3 ,
Enter x_2

x_1	1	0	3	-4	0	8	90
x_5	0	0	2	1	1	2	40
x_2	0	1	1	0	0	3	10
Z	0	0	5	-1	0	25	50

Remove x_5 ,
Enter x_4

x_1	1	0	11	0	4	16	250
x_4	0	0	2	1	1	2	40
x_2	0	1	1	0	0	3	10
Z	0	0	7	0	1	27	90

Solution:

$$z_{\max} = 90 \quad \text{at } (x_1, x_2, x_3, x_4, x_5, x_6) = (250, 10, 0, 40, 0, 0)$$

Question 3

Objective Function:

$$\mathbf{\min} \ z = x_1 + x_2 + x_3$$

Constraints:

- $-x_1 + 2x_2 + x_3 \leq 1$
- $-x_1 + 2x_3 \geq 4$
- $x_1 - x_2 + 2x_3 = 4$
- $x_1, x_2, x_3 \geq 0$ **(non-negative)**

To put this in equation form for Two-Phase simplex: add a slack s_1 to the $\leq$ constraint, subtract a surplus s_2 and add an artificial r_1 to the $\geq$ constraint, and add an artificial r_2 to the $=$ constraint. The initial basic variables are then s_1, r_1 , and r_2 .

Canonical Form (with Slack, Surplus, and Artificial Variables):

Constraints:

- $-x_1 + 2x_2 + x_3 + s_1 = 1$
- $-x_1 + 2x_3 - s_2 + r_1 = 4$
- $x_1 - x_2 + 2x_3 + r_2 = 4$
- $x_1, x_2, x_3, s_1, s_2, r_1, r_2 \geq 0$ **(non-negative)**

Phase I:

Objective Function:

$$\mathbf{\min} \ w = r_1 + r_2$$

Initial BVs: $s_1 = 1, r_1 = 4, r_2 = 4$ (all other variables = 0).

Phase II:

Once Phase I reaches $w = 0$ with all artificials nonbasic, drop the r_i columns and optimize:

$$\min z = x_1 + x_2 + x_3$$

Simplex Algorithm:

	x_1	x_2	x_3	s_1	s_2	r_1	r_2	RHS
s_1	-1	2	1	1	0	0	0	4
r_1	-1	0	2	0	-1	1	0	4
r_2	1	-1	2	0	0	0	1	4
Z	1	1	1	0	0	0	0	0
w	0	0	0	0	0	1	1	0
	0	1	1	0	1	0	0	-8

$w = -r_1 - r_2$

Enter x_3 Remove s_1

	x_3	x_1	x_2	s_1	s_2	r_1	r_2	RHS
x_3	1	2	1	1	0	0	0	1
r_1	1	-4	0	-2	-1	1	0	2
r_2	3	-5	0	-2	0	0	1	2
Z	2	-1	0	-1	0	0	0	-1
w	-4	9	0	4	1	0	0	-11

Enter x_1
Remove r_2

	x_3	x_1	x_2	s_1	s_2	r_1	r_2	RHS
x_3	0	$\frac{1}{3}$	1	$\frac{2}{3}$	0	0	0	$\frac{2}{3}$
r_1	0	$-\frac{7}{3}$	0	$-\frac{2}{3}$	-1	1	$-\frac{1}{3}$	$-\frac{4}{3}$
x_1	1	$-\frac{5}{3}$	0	$-\frac{2}{3}$	0	0	$\frac{1}{3}$	$\frac{2}{3}$
Z	0	$\frac{4}{3}$	0	$\frac{4}{3}$	1	0	$\frac{1}{3}$	$-\frac{11}{3}$
w	0	$\frac{4}{3}$	0	$\frac{4}{3}$	1	0	$\frac{1}{3}$	$-\frac{11}{3}$

Solution:

The problem is **infeasible**. At the end of Phase I, every entry in the w -row (among the variable columns) is nonnegative, so Phase I is optimal, but w is not zeroed out (RHS = $-4/3$, so $w = 4/3 > 0$). An artificial variable remains positive, so there is no feasible solution to the original LP, and Phase II is not needed.

Question 4

Determine two distinct basic feasible solutions at which the optimal value of the objective function is attained.

Objective Function:

$$\min x_1 - 3x_2 - 6x_3$$

Constraints:

- $2x_1 - x_2 + x_3 + x_4 \leq 60$
- $3x_1 + 4x_2 + 2x_3 - 2x_4 \leq 150$
- $x_1, x_2, x_3, x_4 \geq 0$ **(non-negative)**

Both constraints are $\leq$, so I add slack variables s_1 and s_2 to put the problem in canonical form. No artificial variables are needed.

Canonical Form:**Objective Function:**

$$\min x_1 - 3x_2 - 6x_3$$

Constraints:

- $2x_1 - x_2 + x_3 + x_4 + s_1 = 60$
- $3x_1 + 4x_2 + 2x_3 - 2x_4 + s_2 = 150$
- $x_1, x_2, x_3, x_4, s_1, s_2 \geq 0$ **(non-negative)**

Simplex Algorithm:

Initial Simplex Tableau:

	x_1	x_2	x_3	x_4	s_1	s_2	RHS
s_1	2	-1	1	1	1	0	60
s_2	3	4	2	-2	0	1	150
z	1	-3	6	0	0	0	0

Enter x_3 , Exit s_1 $\Rightarrow$

Second Simplex Tableau:

	x_3	x_2	x_4	s_2	RHS
x_3	2	-1	1	1	60
s_2	-1	6	0	-4	30
z	13	-9	0	6	360

Enter x_2 , Exit s_2

Third Simplex Tableau:

	x_3	x_2	x_4	s_2	RHS
x_3	$\frac{11}{2}$	0	1	$\frac{1}{3}$	65
x_2	$-\frac{1}{6}$	1	0	$-\frac{2}{3}$	5
z	$13-\frac{9}{2}$	0	0	3	405

BFS: $z = -405$ @ $(0, 5, 65, 0)$

Enter x_4 , Remove x_3

Fourth Simplex Tableau:

	x_4	x_2	s_2	RHS
x_4	$\frac{11}{2}$	0	3	195
x_2	0	1	0	135
z	...	0	...	405

BFS: $z = -405$ @ $(0, 135, 0, 19.5)$

At the first optimal tableau, a nonbasic variable (x_4) has a zero coefficient in the objective row. That means bringing x_4 into the basis does not change the objective value, so I can pivot on that column to get

a different BFS that is still optimal.

Solution:

Two distinct optimal basic feasible solutions, both with $z = -405$:

- $(x_1, x_2, x_3, x_4) = (0, 5, 65, 0)$
- $(x_1, x_2, x_3, x_4) = (0, 135, 0, 195)$